Figure S1. Pooled analysis for the comparison of the risk for postoperative atrial fibrillation (POAF) after removal of the study with the smallest sample size.

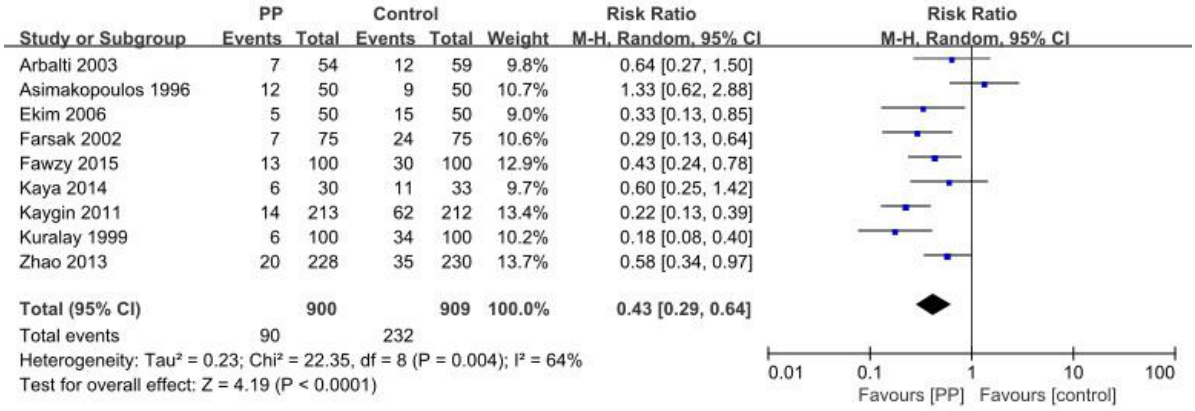

Figure S2. Pooled analysis for the comparison of the risk for postoperative atrial fibrillation (POAF) without preoperative oral β-blockers.

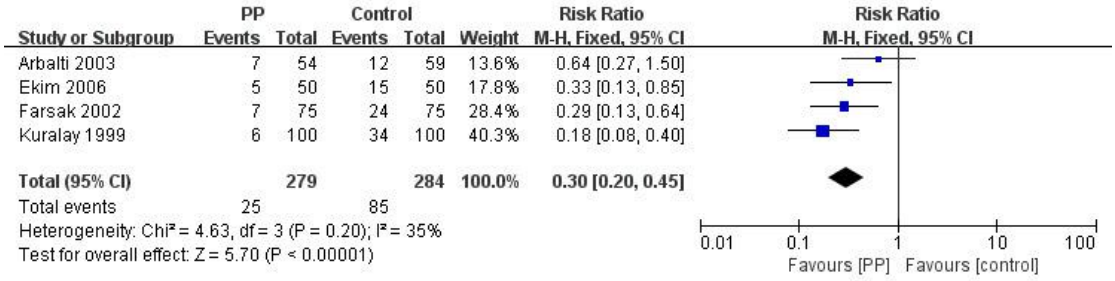

Figure S3. Pooled analysis for the comparison of the risk for postoperative atrial fibrillation (POAF) in Turkey.

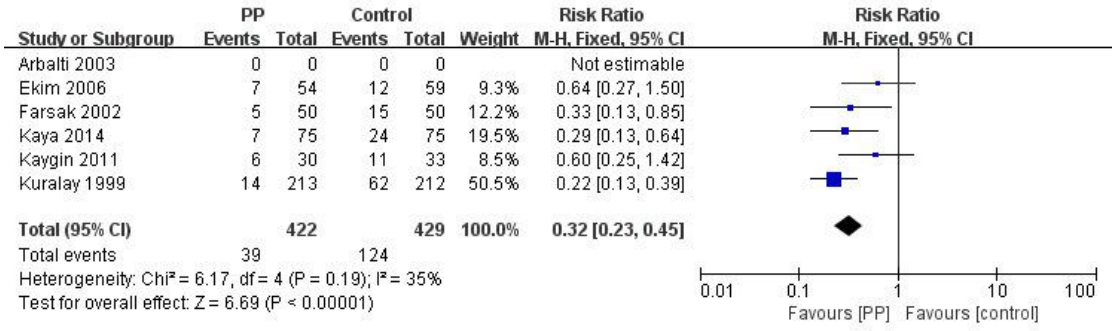

Figure S4. Pooled analysis for the comparison of the risk for postoperative early pericardial effusion.

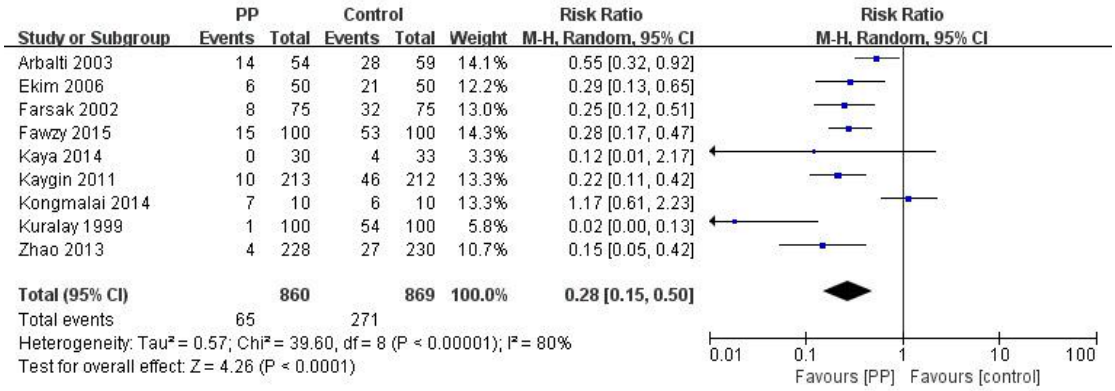

Figure S5. Pooled analysis for the comparison of the risk for postoperative late pericardial effusion.

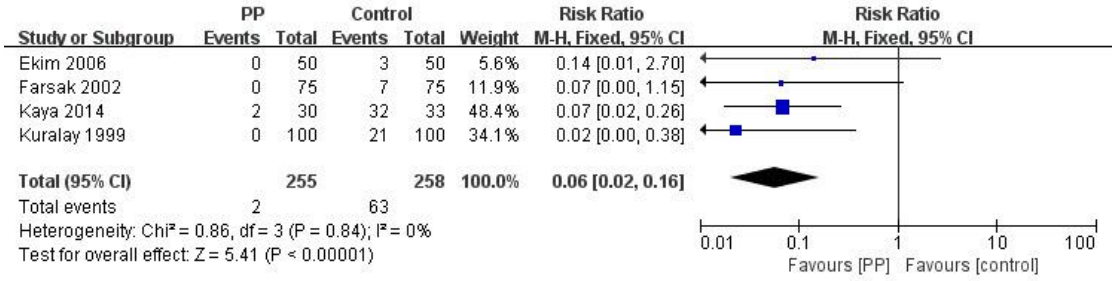

Figure S6. Pooled analysis for the comparison of the risk for postoperative pericardial tamponade.

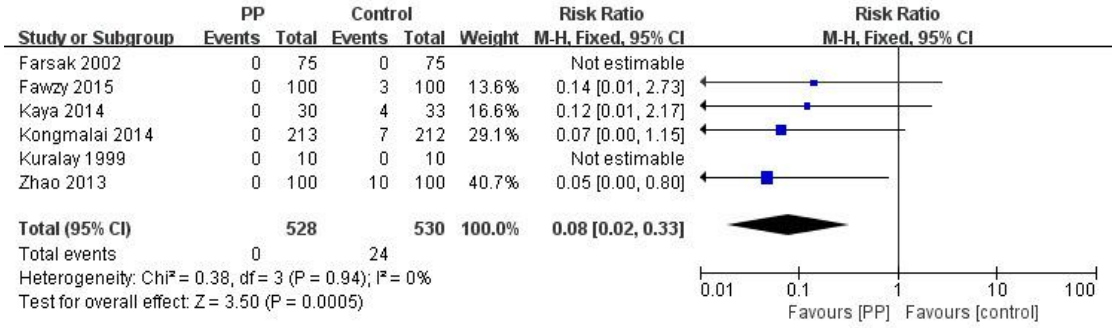

Figure S7. Pooled analysis for the comparison of the risk for postoperative length of stay in intensive care unit (ICU).

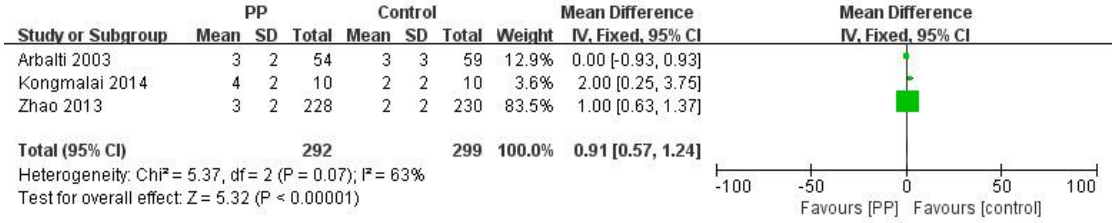

Figure S8. Pooled analysis for the comparison of the risk for postoperative pleural effusion.

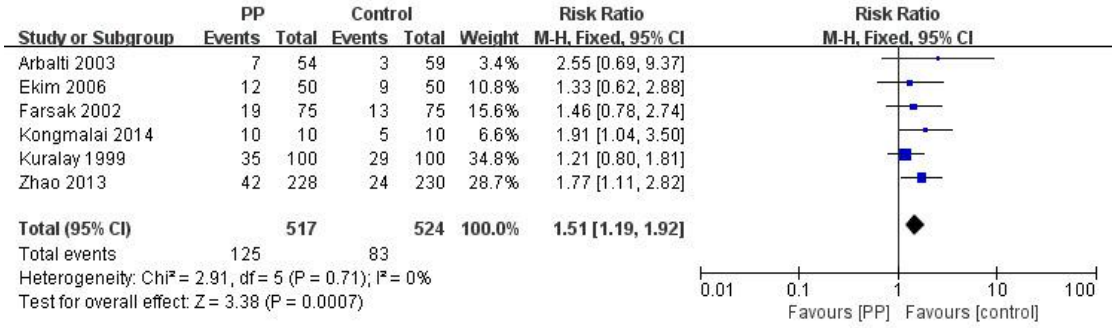

Figure S9. Pooled analysis for the comparison of the risk for postoperative length of hospitalization.

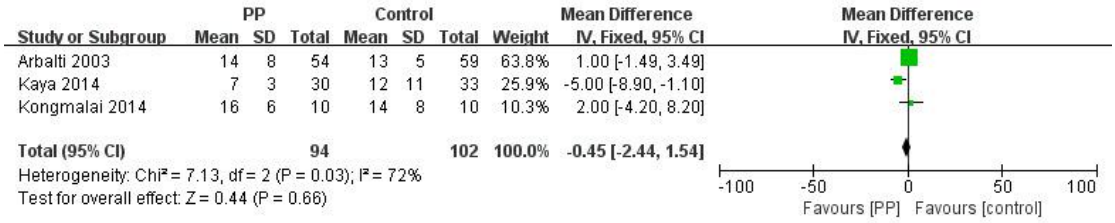

Figure S10.Pooled analysis for the comparison of the risk for postoperative pulmonary complications.

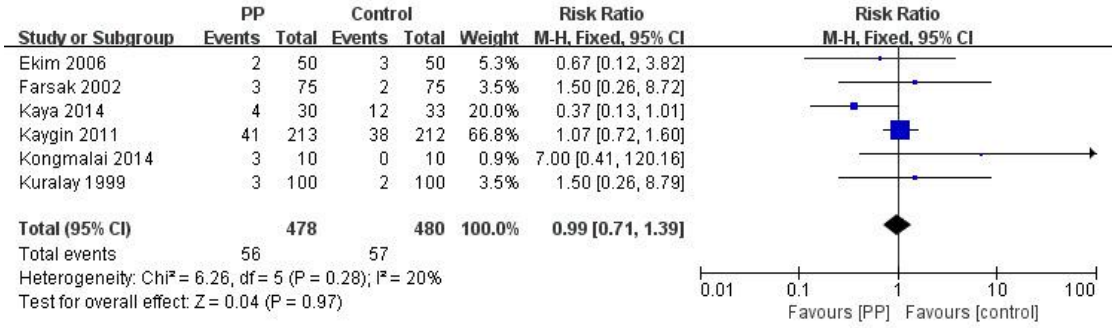

Figure S11. Pooled analysis for the comparison of the risk for postoperative revision for bleeding.

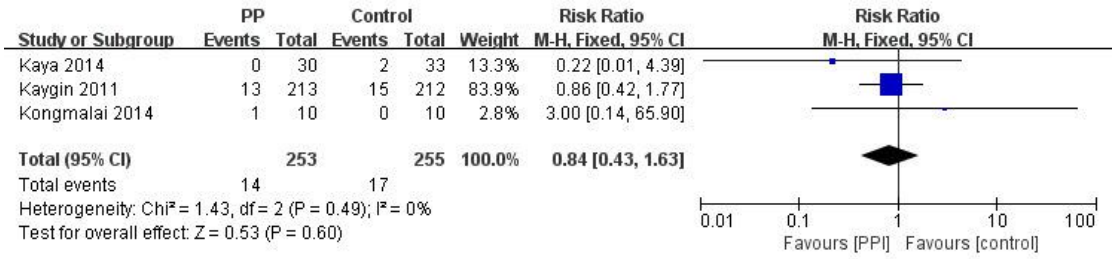

Figure S12. Pooled analysis for the comparison of the risk for postoperative intra aortic balloon pump (IABP) usage.

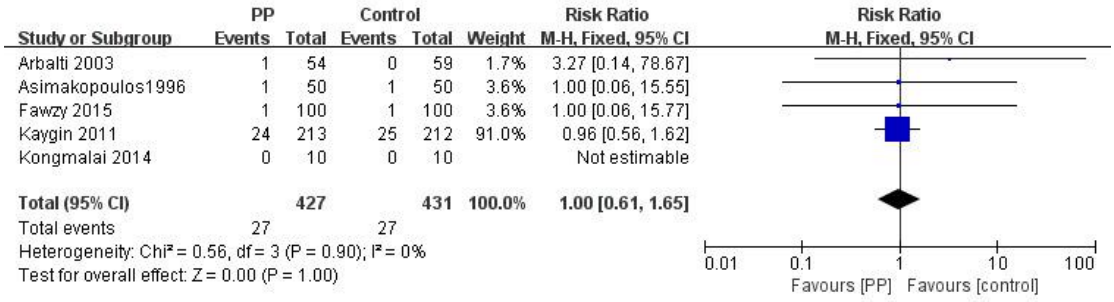

Figure S13. Pooled analysis for the comparison of the risk for postoperative death.

Table S1. Main postoperative data from random controlled trials included in the meta-analysis

StudyID	POAF		Early pericardial effusion		Late pericardial effusion		Pulmonary complication		Pericardial tamponade	
	PP	control	PP	control	PP	control	PP	control	PP	control
Kaya 2014[34]	6	11	0	4	2	32	4	12	0	4
Kaygin 2011[33]	14	62	10	46	NR	NR	41	38	0	7
Fawzy 2015[35]	13	30	15	53	NR	NR	NR	NR	0	3
Zhao 2013[29]	20	35	4	27	NR	NR	NR	NR	3	13
Kuralay 1999[18]	6	34	1	54	0	21	3	2	0	10
Asimakopoulos 1996[32]	12	9	NR	NR	NR	NR	NR	NR	NR	NR
Ekim 2006[20]	5	15	6	21	0	3	2	3	NR	NR
Farsak 2002[19]	7	24	8	32	0	7	3	2	0	0
Kongmalai 2014[22]	4	4	7	6	NR	NR	3	0	0	0
Arbalti 2003[21]	7	12	14	28	NR	NR	NR	NR	NR	NR

ICU stay，Days		Hospitalization time, days		IABP usage		Mortality		Revision for bleeding		Pleural effusion	
PP	control	PP	control	PP	control	PP	control	PP	control	PP	control
NR	NR	7±3	12±11	NR	NR	0	2	0	2	NR	NR
NR	NR	NR	NR	24	25	NR	NR	13	15	NR	NR
NR	NR	8	9	1	1	NR	NR	NR	NR	NR	NR
3±2	2±2	NR	NR	NR	NR	NR	NR	NR	NR	42	24
NR	NR	7	8	NR	NR	NR	NR	NR	NR	35	29
NR	NR	NR	NR	1	1	1	1	NR	NR	NR	NR
NR	NR	NR	NR	NR	NR	NR	NR	NR	NR	12	9
NR	NR	NR	NR	NR	NR	NR	NR	NR	NR	19	13
4±2	2±2	16±6	14±8	0	0	NR	NR	1	0	10	5
3±2	3±3	14±8	13±5	1	0	NR	NR	NR	NR	7	3

CABG,coronary artery bypass grafting; POAF,postoperative atrial fibrillation; ICU,intensive care unit; RCT,randomized controlled trial; NR, not reported; CPB,cardiopulmonary bypass; IABP,intra-aortic ballon pump.

Table S2. The actual mode of PP and the use of posterior pericardial drains

StudylD	the use of PP	the use of posterior pericardial drains
Kaya 2014[34]	a 4-cm longitudinal incision was made parallel and posterior to the left phrenic nerve, extending from the left inferior pulmonary vein to the diaphragm	A straight tube was placed in the anterior mediastinum and an angled tube was placed in the left hemithorax in all patients. The left pleural cavity was routinely opened
Kaygin2011[33]	a 4-cm longitudinal incision was made parallel and posterior to the left phrenic nerve, extending from the left inferior pulmonary vein to the diaphragm	Two chest tubes (left pleural cavity and anterior mediastinum) were inserted into the pericardium.
Fawzy2015[35]	a longitudinal incision, 4-cm long and 2-cm width, was made parallel and posterior to the left phrenic nerve, extending from the left inferior pulmonary vein to the diaphragm	Two chest tubes (one in the left pleural cavity and the second in the anterior mediastinum) were inserted a 28 F rubber drain tube was placed along the right atrium in the pericardium and a 32 F silicone tube was placed retrosternally in the anterior mediastinum overlying the heart.The tubes were connected using a T-shaped,negative-pressure, drainage device.
Zhao 2013[29]	The incision was made into the pleural cavity by noncontinuous electrocautery, parallel and posterior to the phrenic nerve and extending from the left inferior pulmonary vein to the diaphragm	Two chest tubes (one in the left pleural cavity and the other in anterior mediastinum) were inserted
Kuralay 1999[18]	Longitudinal incision was made parallel and posterior to the left phrenic nerve, extending from the left inferior pulmonary vein to the diaphragm	
Asimakopoulos 1997[32]	a 4 cm posterior pericardial incision below the left inferior pulmonary vein parallel and posterior to the phrenic nerve	
Ekim 2006[20]	A 4-cm longitudinal incision was made parallel and 1.5 cm posterior to the phrenic nerve, extending from the left inferior pulmonary vein to the diaphragm	
Farsak2002[19]	4-cm longitudinal incision was made parallel and posterior to the left phrenic nerve, extending from the left inferior pulmonary vein to the diaphragm	
Kongmalai 2014[22]	a 4-cm circular incision was made in parallel and posterior to the left phrenic nerve, extending from the left inferior pulmonary vein to the diaphragm	
Arbalti 2003[21]	a longitudinal incision was made parallel and posterior to the phrenic nerve, extending from the left inferior pulmonary vein to the diaphragm	

Anterior mediastinal and left pleural drains were routinely placed with suction at 10 mm Hg

Two chest tubes were placed, one in the left pleural cavity and the other in the anterior mediastinum

Two chest tubes (one in the left pleural cavity and the other in anterior mediastinum)

One drainage tube was inserted into the left thorax and one or two into the anterior mediastinum, without suction

Two chest drains were inserted, one in the left pleural cavity and the other in the antermr mediastinum

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PP,Posterior pericardiotomy